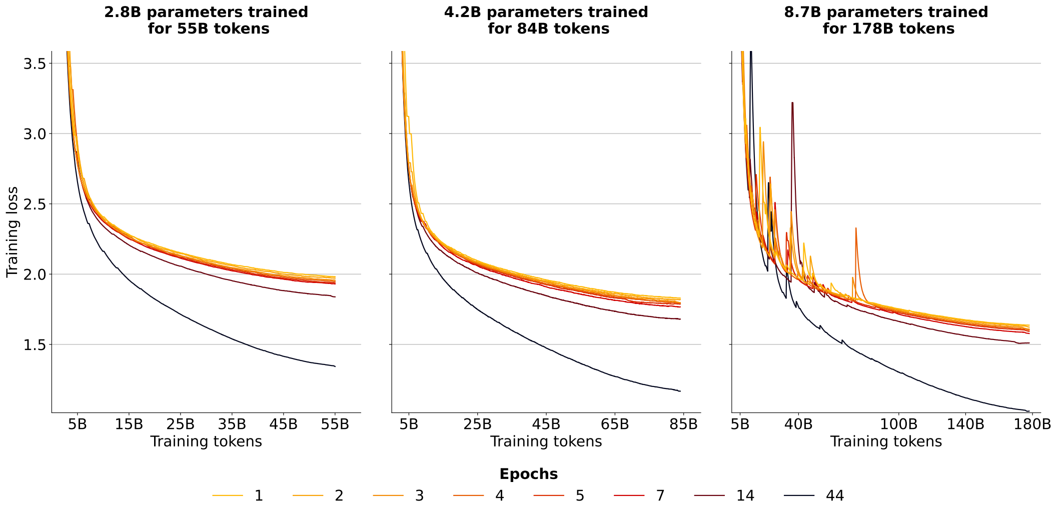

Figure 16: **Training loss for models trained on OSCAR smoothed with exponential moving average smoothing and a weight of 0.999.** Models trained on fewer unique tokens (more epochs) have better training loss as they overfit.

J Validation Loss by Epoch

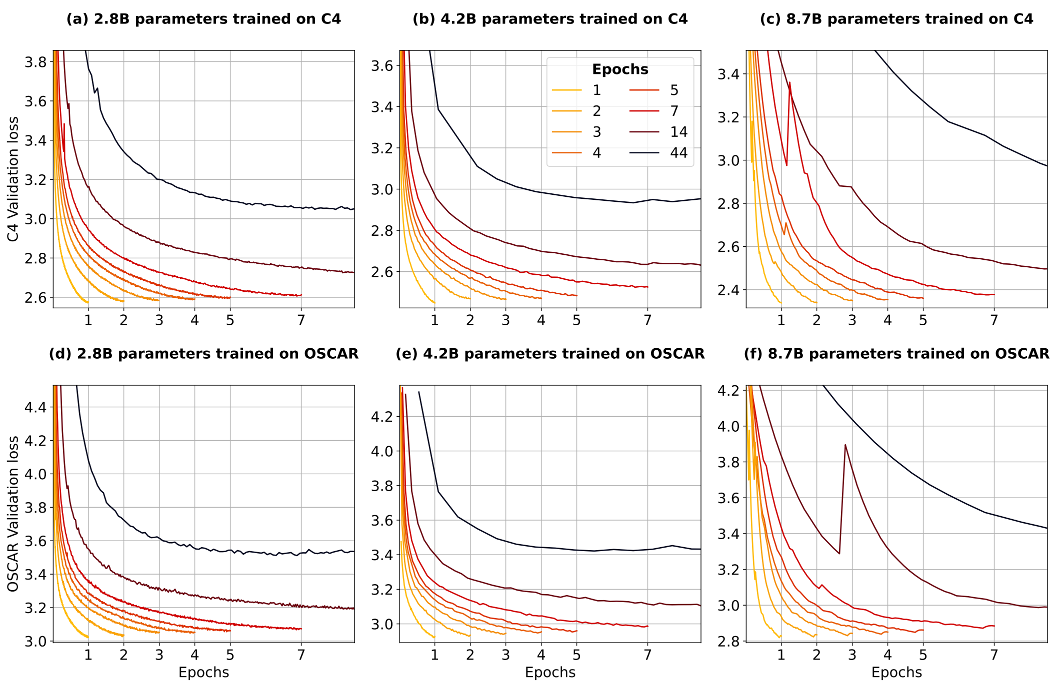

Figure 17: **Validation loss during training visualized by epochs.** Loss progresses smoothly throughout training. There are temporary spikes for 8.7 billion parameter models, commonly at the start of a new epoch.

Taylor et al. [108] decided to early-stop pre-training of the Galactica models due to a small increase in validation loss at the start of the fifth epoch. In Figure 17 we plot the validation loss curves of our isoFLOP models as a function of epochs. We do find small increases in validation loss when models enter a new epoch. For example, upon entering the third and fourth epoch, the 7-epoch 8.7 billion parameter OSCAR model shows loss spikes. However, these are temporary and loss continues to go down smoothly thereafter. Thus, we hypothesize that the Galactica models could have attained better performance by continuing pre-training beyond the loss spike experienced at the beginning of the fifth epoch.

K Evaluation Details

Table 2: **Setup for computing validation loss during training.** At every *Evaluation Interval*, loss is computed on *Evaluation Tokens* many tokens from the validation set. The evaluation tokens vary with the interval, i.e. the evaluation tokens at 100 steps are not the same as at 200 steps. However, the tokens do not vary across data budgets for the same FLOP budget (Figure 4). For example, $N = 2.8$ billion parameter models with $D_C = 55$ billion tokens are evaluated on the same data as models with $D_C = 28$ billion tokens at each evaluation interval.

FLOP budget	Parameters	Evaluation Interval	Evaluation Tokens
9.3×10^{20}	2.8B	100	105 million
2.1×10^{21}	4.2B	1000	105 million
9.3×10^{21}	8.7B	1000	2.1 million

Loss evaluation For all models trained on C4, the final test loss is computed on the same 210 million tokens from the C4 validation set after training. For held-out evaluation during training, such as in Figure 4, the configurations are displayed in Table 2. The small number of evaluation tokens for the 8.7 billion parameter models likely contributes to the loss spikes for 8.7 billion parameter models seen in Figure 4. Thus, we smooth the validation loss curves of 8.7 billion parameter models with exponential moving average smoothing and a weight of 0.85. For training on OSCAR, configurations are the same, however, the validation split used is a held-out part from the OSCAR training split, as there is no official validation split for OSCAR. All training loss curves for C4 and OSCAR models are smoothed with exponential moving average smoothing and a weight of 0.999.

Downstream evaluation We provide statistics of all downstream evaluation datasets in Table 3. We use the evaluation-harness frameworks from BigScience and EleutherAI [32] to evaluate models on 19 evaluation datasets. For each dataset, a maximum of 3000 samples are evaluated with 0,1,2,3,4 and 5 few-shots [15] to produce six scores which are then averaged. We normalize scores to range from the random baseline of each task to 1 and report them as percentages. For example, if random guessing produces 50% accuracy and the maximum accuracy possible is 100%, then a raw accuracy of 55% would be normalized to 10%, and a raw accuracy of 45% would be normalized to -10% since it is worse than random. This is done to give all tasks the same weight. Otherwise average performance would heavily depend on generative tasks, where the random baselines are 0. Prompts are sourced from GPT-3 [15] and PromptSource [5] and detailed in Appendix T. We note that our evaluation is in no means comprehensive and a larger benchmarking would be helpful [102, 73]. However, by training five seeds for most models benchmarked, always averaging 0-5 fewshots, and ensuring maximum data overlap for repeated data (§4) we significantly reduce uncertainty.

Table 3: **Downstream evaluation datasets.** We evaluate on 19 datasets: The first 14 are evaluated using accuracy (ANLI counted as three), the next 4 using ROUGE-2 f-measure [59] and bAbI using exact match.

Dataset	Split(s)	Samples	Baseline	URL
ANLI [79]	dev_r1,2,3	3000	33.3	hf.co/datasets/anli
ARC-Easy [22]	test	1172	25.0	hf.co/datasets/ai2_arc
ARC-Challenge [22]	test	2376	25.0	hf.co/datasets/ai2_arc
BoolQ [21]	validation	3270	50.0	hf.co/datasets/boolq
CB [25]	validation	56	33.3	hf.co/datasets/super_glue
Copa [92]	validation	100	50.0	hf.co/datasets/super_glue
HellaSwag [129]	test	10003	25.0	hf.co/datasets/hellaswag
PiQA [12]	validation	1838	50.0	hf.co/datasets/piqa
RTE [24, 114]	validation	277	50.0	hf.co/datasets/super_glue
SciQ [120]	test	1000	25.0	hf.co/datasets/sciq
StoryCloze 2016 [69]	test	1871	25.0	hf.co/datasets/story_cloze
WinoGrande XL [93]	test	1267	50.0	hf.co/datasets/winogrande
E2E NLG [29]	test	4693	0.0	hf.co/datasets/e2e_nlg_cleaned
XSUM [77, 34]	test	11334	0.0	hf.co/datasets/GEM/xsum
WebNLG EN [17, 34]	test	5150	0.0	hf.co/datasets/GEM/web_nlg
WikiLingua EN [53, 34]	sampled_test	3000	0.0	hf.co/datasets/GEM/wiki_lingua
bAbI [122]	test	19000	0.0	hf.co/datasets/Muennighoff/babi

L Downstream Repetition Results

In Tables 4-9 we report downstream results of all models trained on C4 [90] and OSCAR [83] according to the configurations in Figure 4. All scores are from the final checkpoints at the end of training. OSCAR is a noisier dataset than C4 due to less filtering, thus models trained on C4 generally perform better. Notably, models trained on C4 completely fail on bAbI [122], while OSCAR models are able to perform better than random. This is likely due to code data being present in OSCAR, which enables state-tracking capabilities like for code augmented models in §7. For C4 the creators strictly removed all data that resembles code [90]. There are no significant differences between models trained for a single epoch and models trained for up to 4 epochs. Even models trained for more epochs (and thus on less unique data) have similar performance.

Table 4: **Results for 2.8B parameter models trained on repeated data on C4 for 55B total tokens.** Scores are normalized averages of 0-5 few-shots and reported as percentages. We report mean/std. err. across five different models, each trained with a different random seed.

Data Budget	55B	28B	18B	14B	11B	9B	4B	1.25B
Epochs	1	2	3	4	5	7	14	44
ANLI R1	0.4 ± 1.6	0.7 ± 0.8	0.3 ± 0.5	-0.3 ± 1.8	0.4 ± 1.8	0.4 ± 0.7	0.0 ± 0.9	-0.6 ± 0.6
ANLI R2	0.9 ± 0.4	1.4 ± 0.8	0.8 ± 0.8	1.1 ± 0.7	0.5 ± 0.7	0.6 ± 1.0	1.1 ± 1.1	2.7 ± 1.6
ANLI R3	1.7 ± 0.5	1.2 ± 0.4	0.4 ± 0.5	1.9 ± 0.7	0.6 ± 1.0	0.8 ± 0.8	1.7 ± 0.7	0.7 ± 1.7
ARC-Challenge	1.6 ± 1.0	0.9 ± 0.5	1.2 ± 0.6	1.1 ± 0.6	1.1 ± 1.2	1.3 ± 0.5	0.3 ± 0.6	-2.9 ± 1.0
ARC-Easy	44.5 ± 0.5	44.9 ± 0.4	44.7 ± 0.7	44.3 ± 0.4	44.0 ± 0.5	44.2 ± 0.9	41.4 ± 0.2	28.9 ± 0.7
BoolQ	18.8 ± 3.4	16.2 ± 5.2	16.1 ± 2.7	19.7 ± 1.8	15.0 ± 3.8	16.9 ± 3.2	13.1 ± 4.9	-2.1 ± 4.7
CB	20.0 ± 4.7	17.4 ± 6.4	14.6 ± 5.1	17.5 ± 4.2	12.3 ± 12.2	14.4 ± 7.5	21.6 ± 8.4	21.3 ± 5.6
COPA	49.7 ± 3.5	50.3 ± 3.4	49.9 ± 2.3	50.1 ± 2.5	50.9 ± 1.2	48.1 ± 2.4	43.5 ± 3.1	33.3 ± 1.9
HellaSwag	24.7 ± 0.3	24.6 ± 0.2	24.3 ± 0.1	24.3 ± 0.0	24.3 ± 0.3	24.1 ± 0.1	22.8 ± 0.2	16.7 ± 0.4
PiQA	47.9 ± 0.6	47.6 ± 0.8	47.3 ± 0.3	47.6 ± 0.6	47.6 ± 0.7	47.0 ± 0.2	45.6 ± 0.5	37.0 ± 0.4
RTE	5.1 ± 4.0	2.5 ± 4.5	8.4 ± 2.6	6.0 ± 2.5	5.1 ± 1.6	2.3 ± 3.9	7.8 ± 2.5	2.6 ± 4.3
SciQ	83.2 ± 0.6	82.5 ± 0.6	82.7 ± 1.1	81.9 ± 0.6	81.9 ± 0.8	81.6 ± 0.9	78.5 ± 1.1	59.3 ± 1.6
StoryCloze 2016	58.7 ± 0.2	58.7 ± 0.5	58.5 ± 0.3	58.3 ± 0.3	58.5 ± 0.6	58.4 ± 0.3	56.7 ± 0.5	52.0 ± 0.6
WinoGrande XL	11.6 ± 0.8	10.8 ± 1.1	10.9 ± 1.3	10.6 ± 0.5	11.1 ± 0.9	10.6 ± 0.9	6.4 ± 1.3	2.9 ± 1.3
E2E NLG	17.0 ± 1.4	17.7 ± 0.5	17.0 ± 1.2	16.9 ± 1.1	15.1 ± 2.3	13.3 ± 2.2	14.9 ± 0.9	9.8 ± 0.9
XSUM	2.4 ± 0.1	2.4 ± 0.1	2.5 ± 0.1	2.3 ± 0.2	2.4 ± 0.1	2.4 ± 0.1	2.1 ± 0.1	1.6 ± 0.1
WebNLG EN	5.3 ± 0.1	5.5 ± 0.2	5.4 ± 0.1	5.4 ± 0.1	5.1 ± 0.1	5.4 ± 0.2	5.1 ± 0.3	2.9 ± 0.2
WikiLingua EN	3.0 ± 0.1	3.1 ± 0.1	2.9 ± 0.1	2.9 ± 0.3	2.9 ± 0.2	2.9 ± 0.1	2.6 ± 0.1	2.0 ± 0.2
bAbI	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
Average	20.9 ± 0.4	20.4 ± 0.3	20.4 ± 0.2	20.6 ± 0.2	19.9 ± 0.9	19.7 ± 0.2	19.2 ± 0.5	14.1 ± 0.4